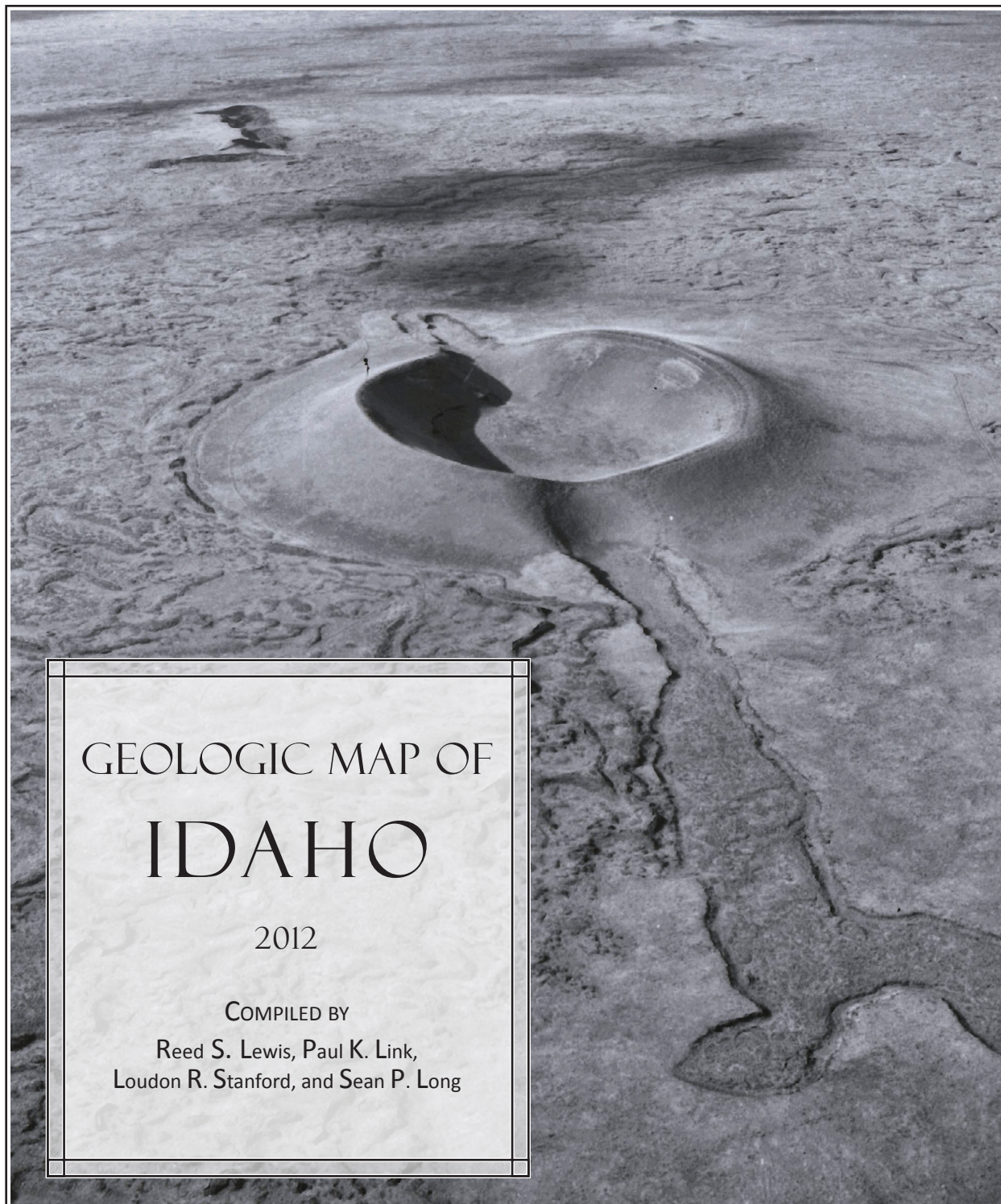




GEOLOGIC MAP OF
IDAHO

2012

COMPILED BY

Reed S. Lewis, Paul K. Link,
Loudon R. Stanford, and Sean P. Long

Geologic Map of Idaho

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University of Idaho
Moscow, Idaho 83843-3014

Geologic Map 9
2012

*Front cover photo: Oblique aerial
view of Sand Butte, a maar crater;
northeast of Richfield, Lincoln County.
Photograph Ronald Greeley.*

Geologic Map Idaho

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2012

INTRODUCTION

The *Geologic Map of Idaho* brings together the extensive mapping and associated research released since the previous statewide compilation by Bond (1978). The geology is compiled from more than ninety map sources (Figure 1). Mapping from the 1980s includes work from the U.S. Geological Survey Conterminous U.S. Mineral Appraisal Program (Worl and others, 1991; Fisher and others, 1992). Mapping from the 1990s includes work by the U.S. Geological Survey during mineral assessments of the Payette and Salmon National forests (Evans and Green, 2003; Lund, 2004). In the late 1990s, the Idaho Geological Survey began completing the mapping within 30' x 60' quadrangles, mostly at a scale of 1:24,000, with funding in part by the U.S. Geological Survey Statemap Program (e.g., Lewis and others 2001, 2002, 2005, 2008; Kauffman and others, 2005, 2009). Since the late 1990s, about 30 percent of the state has been mapped, and this work is ongoing. Numerous thesis projects during the past three decades have also provided local detail.

Map units from the various sources were condensed to 74 units statewide, and major faults were identified. Compilation was at 1:500,000 scale. R.S. Lewis compiled the northern and western parts of the state. P.K. Link initially compiled the eastern and southeastern parts and was later assisted by S.P. Long. County geologic maps were derived from this compilation for the *Digital Atlas of Idaho* (Link and Lewis, 2002). Following the county map project, the statewide compilation was updated, scanned, and digitized at the Idaho Geological Survey. A 1:750,000-scale base derived from 10-meter digital elevation models was prepared for the new map. All surficial unit contacts were modified by L.R. Stanford to reflect the detail of the shaded-relief base map. In addition to the map, representative stratigraphic columns were constructed for six areas of the state (Figures 2-6).

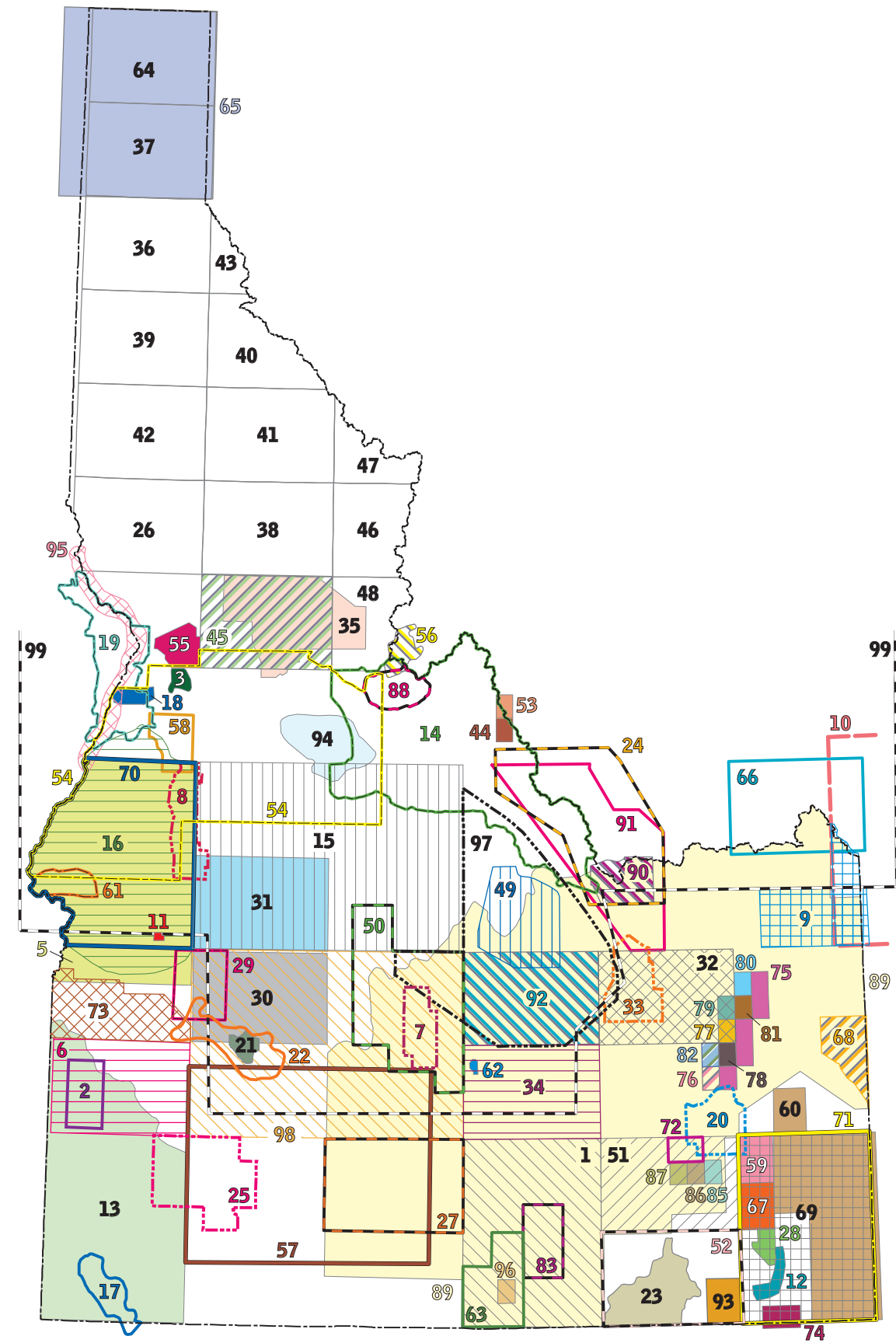
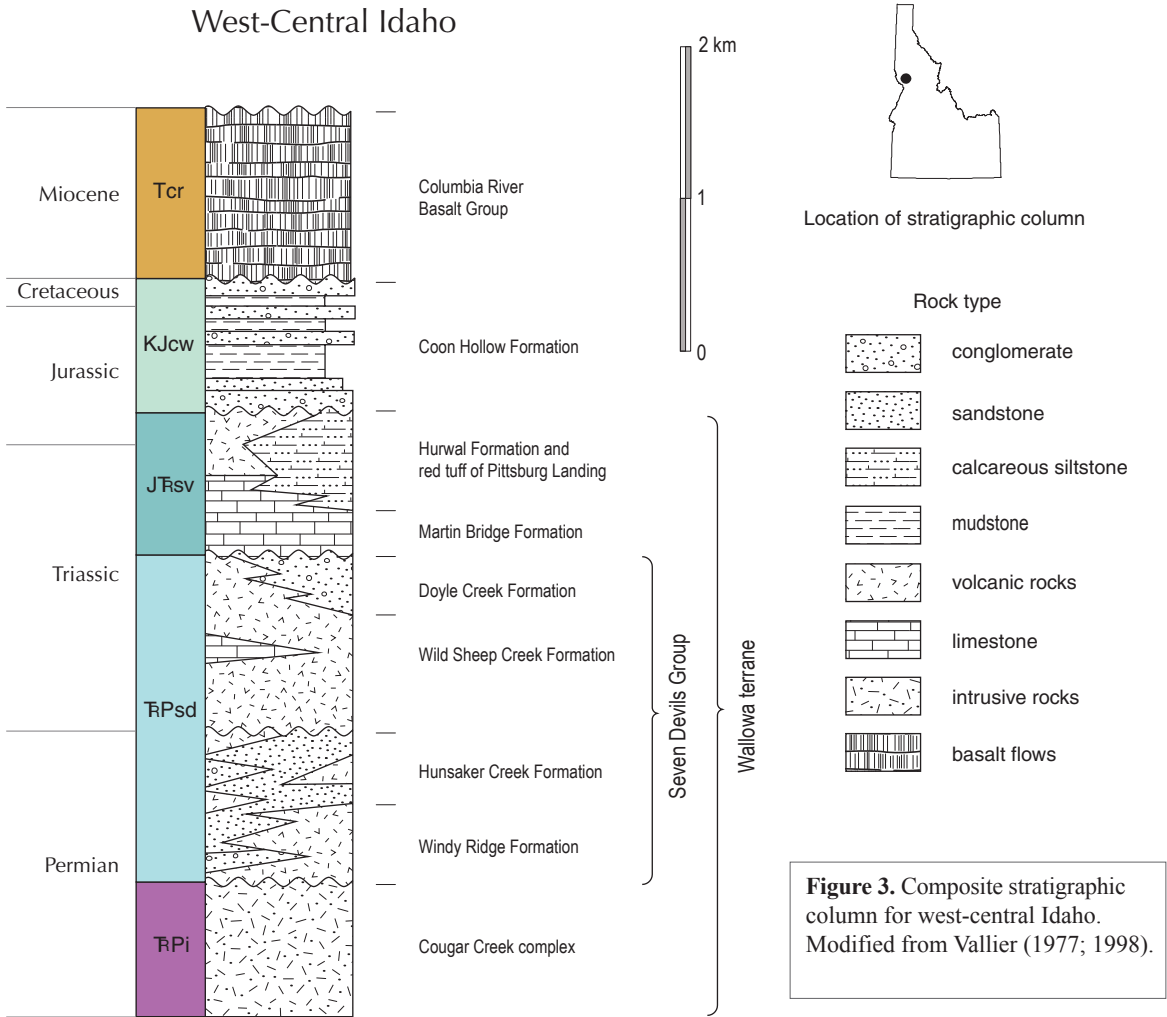
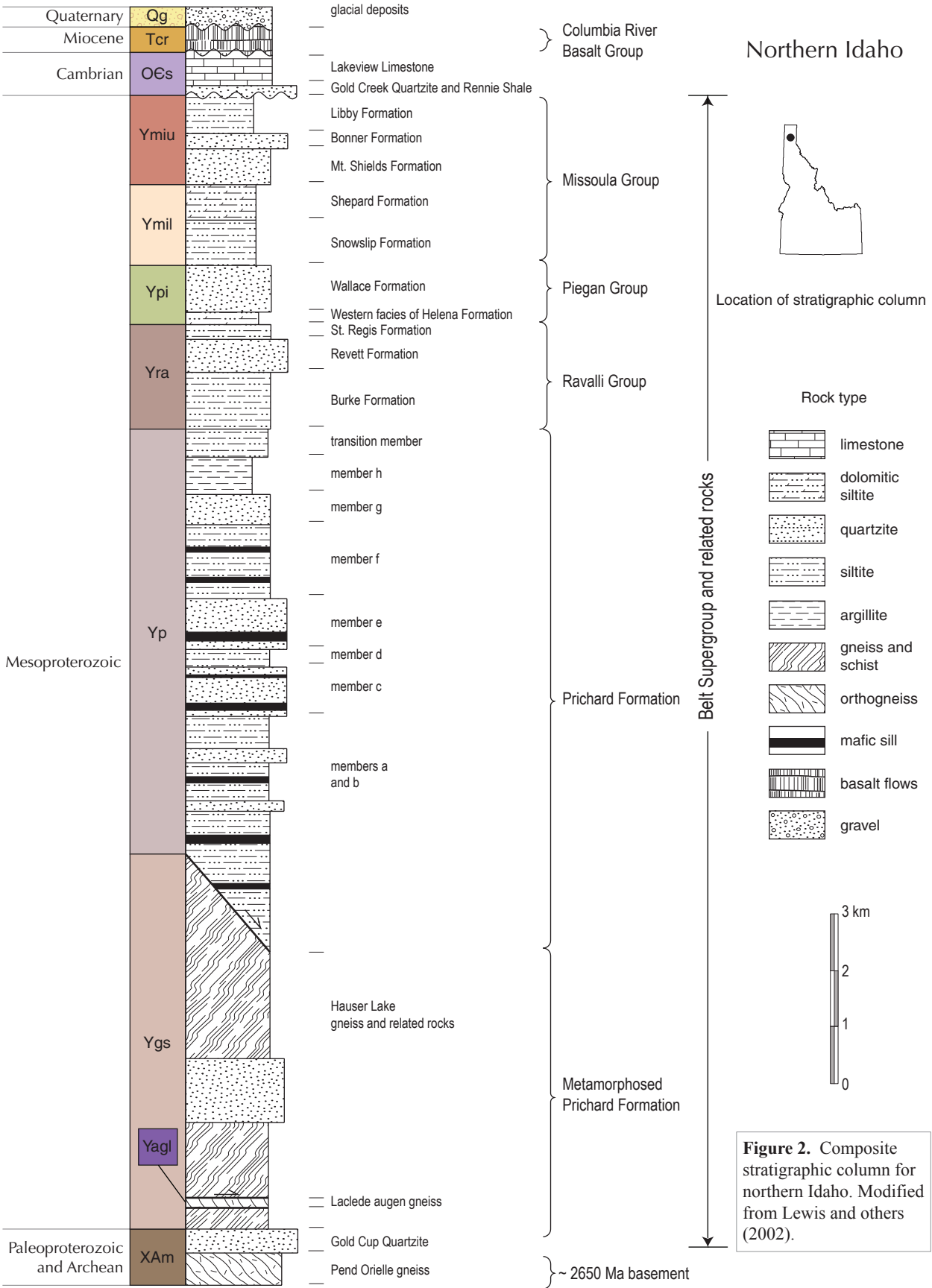
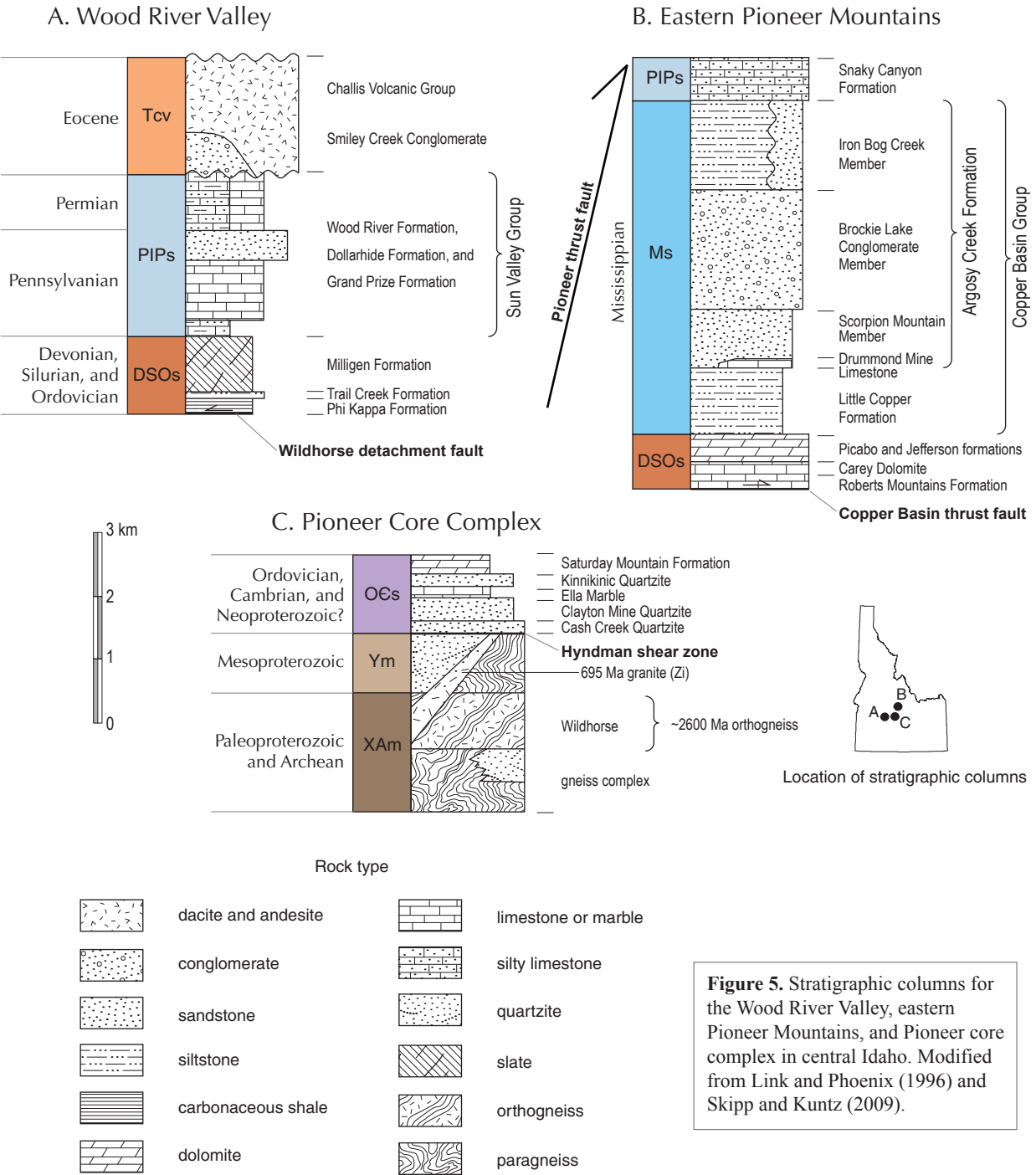
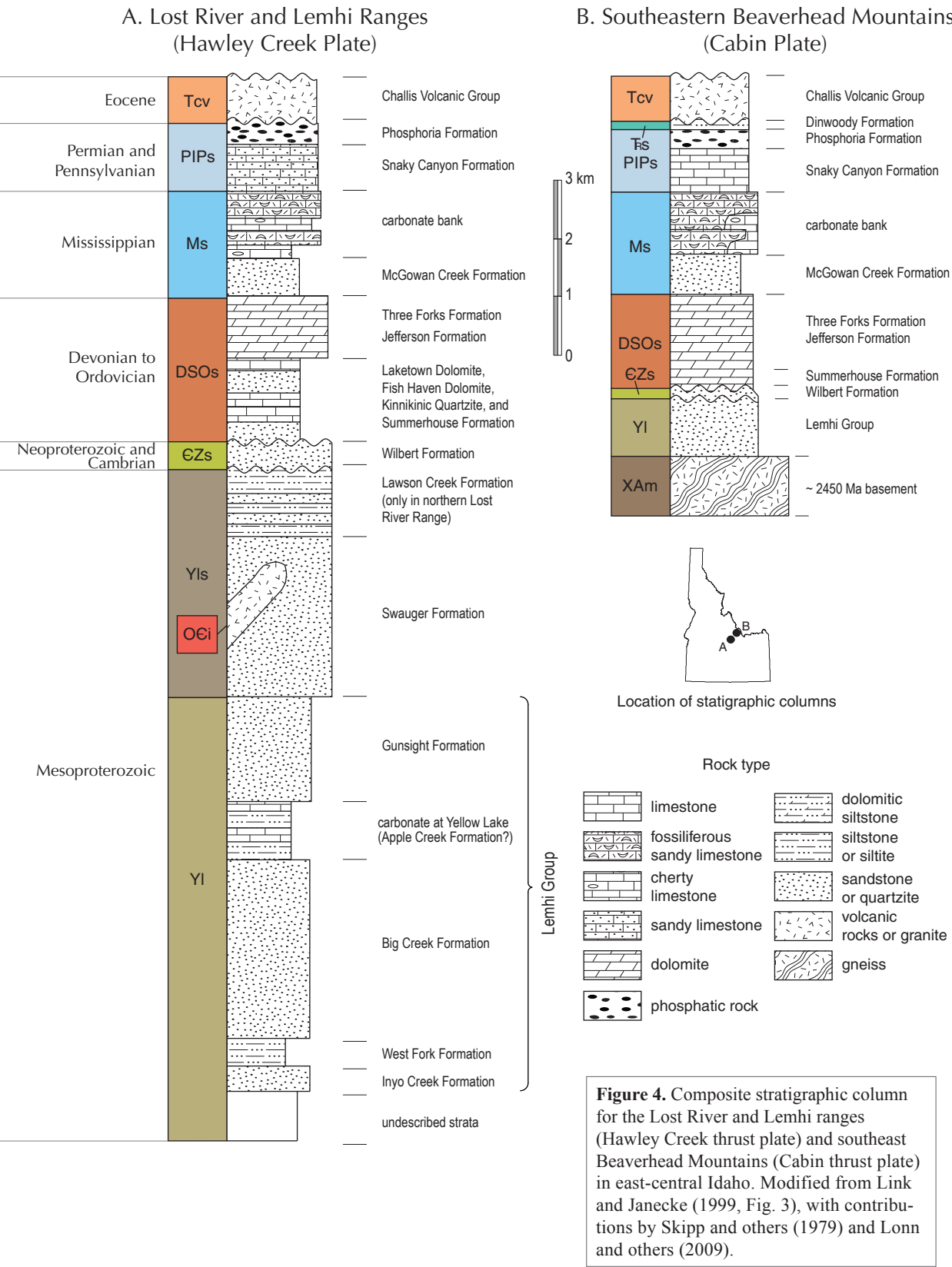


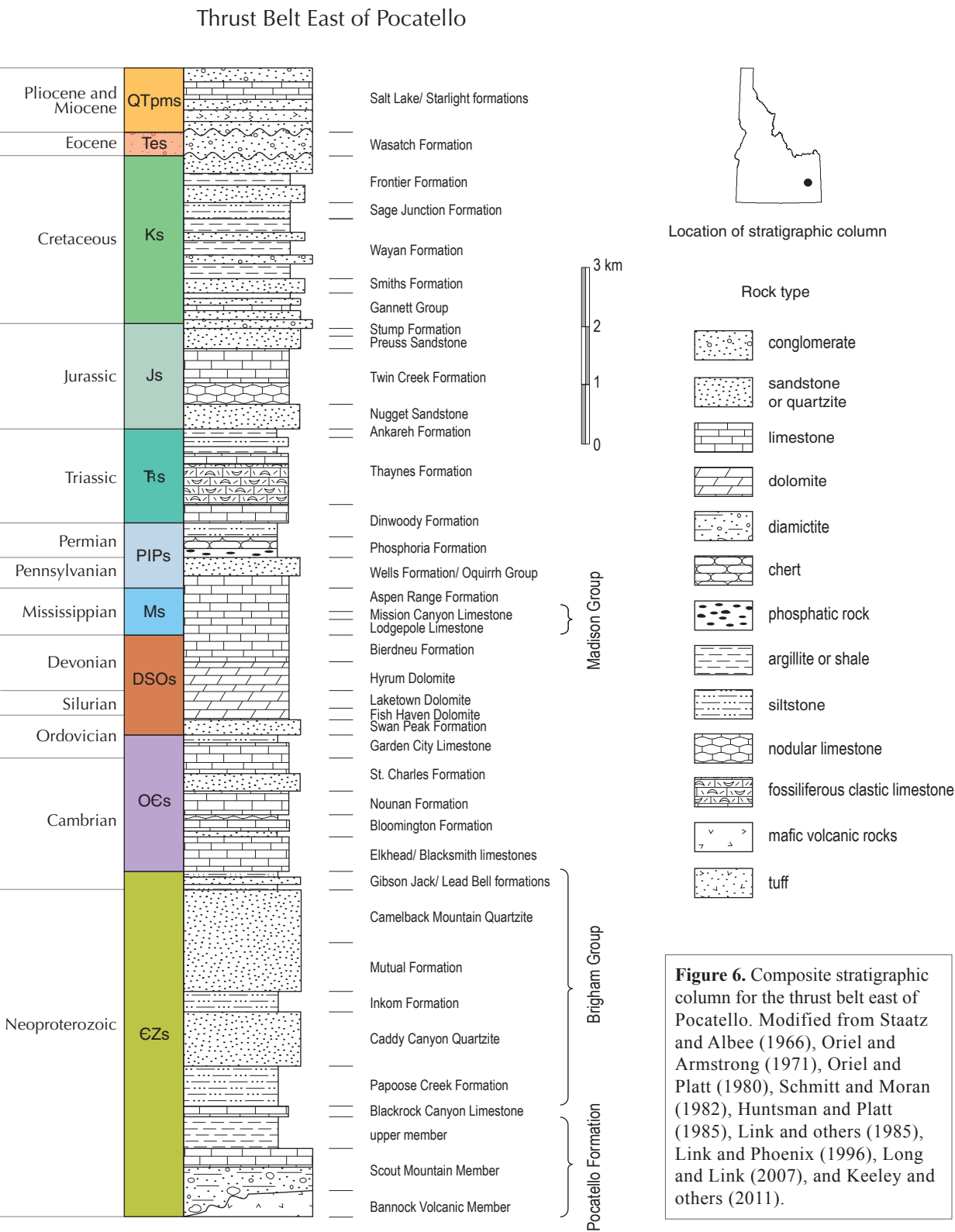
Figure 1. Index map showing sources of geologic mapping.

Index sources of geologic mapping (Figure 1 continued).

1. Armstrong and others, 1978	51. Link and Stanford, 1999
2. Benford and others, 2010	52. Long and Link, 2007
3. Blake, 1991	53. Lonn and others, 2008
4. Bond, 1978 (entire state)	54. Lund, 2004
5. Bond and others, 2011	55. Lund and others, 1993
6. Bonnichsen and Godchaux, 2006	56. Lund and others, 1983
7. Breckenridge and Othberg, 2006b	57. Malde and others, 1963
8. Breckenridge and Othberg, 2006a	58. Manduca and others, 1993
9. Christiansen, 2001	59. Mansfield, 1929
10. Christiansen and Wahl, 1999	60. Mansfield and Girty, 1927
11. Clemens, 1990	61. McIntyre, 1976
12. Cooley and Pederson, 2010	62. Michalek, 2010
13. Ekren and others, 1981	63. Miller, 1983
14. Evans and Green, 2003	64. Miller and Burmester, 2004
15. Fisher and others, 1992	65. Miller and others, 1999
16. Fitzgerald, 1982	66. O'Neill and Christiansen, 2004
17. Foord and others, 1987	67. Oriel, 1968
18. Gray and Oldow, 2005	68. Oriel and Moore, 1985
19. Gualtieri and Simmons, 2007	69. Oriel and Platt, 1980
20. Hladky and others, 1992	70. Othberg, 1982
21. Howard and others, 1982	71. Othberg, 1984
22. Howard and Shervais, 1973	72. Othberg, 2002
23. Hurlow, 2004	73. Othberg and Stanford, 1992
24. Janecke and others, 2001	74. Otto and others, 2005
25. Jenks and others, 1998	75. Phillips and others, 2009
26. Kauffman and others, 2009	76. Phillips and Welhan, 2011a
27. Kauffman and others, 2005	77. Phillips and Welhan, 2011b
28. Keeley and others, 2011	78. Phillips and Welhan, 2011c
29. Kiilsgaard and others, 1997	79. Phillips and Welhan, 2011d
30. Kiilsgaard and others, 2001	80. Phillips and Welhan, 2011e
31. Kiilsgaard and others, 2006	81. Phillips and Welhan, 2011f
32. Kuntz, 2003	82. Phillips and Welhan, 2011g
33. Kuntz and others, 2003	83. Pierce and others, 1983
34. Kuntz and others, 2007	84. USGS, 2006 (entire state)
35. Lewis and others, 1990	85. Riesterer and others, 2000
36. Lewis and others, 2002	86. Rodgers and others, 2006
37. Lewis and others, 2008	87. Rodgers and Othberg, 1999
38. Lewis and others, 2007a	88. Schmidt and others, 1994
39. Lewis and others, 2001	89. Scott, 1982
40. Lewis and others, 1999	90. Skipp, 1984
41. Lewis and others, 2007b	91. Skipp, 1985
42. Lewis and others, 2005	92. Skipp and Kuntz, 2009
43. Lewis and Derkey, 1999	93. Steely and others, 2005
44. Lewis and others, 2009	94. Stewart and others, 2010
45. Lewis and Stanford, 2002a	95. Vallier, 1998
46. Lewis and Stanford, 2002c	96. Waag and others, 1993
47. Lewis and Stanford, 2002d	97. Wilson and Skipp, 1994
48. Lewis and Stanford, 2002b	98. Worl and others, 1991
49. Link and Janecke, 1999	99. Zientek and others, 2005
50. Link and others, 1995	







GEOCHRONOLOGY

Advances in geochronologic techniques since 1978 have greatly improved our understanding of the geologic history of Idaho. Below is a list of geochronologic studies that were particularly helpful in constructing this map.

Lake Bonneville

Janecke and Oaks, 2011
Oviatt, 1997

Active faulting

Thackray and others, 2009

Snake River Plain volcanism

Kuntz and others, 2007
Morgan and McIntosh, 2005
Pierce and Morgan, 1992

Oligocene volcanism and plutonism

Kauffman and others, 2006
Miller and others, 2002
Norman and others, 1986

Idaho batholith and Challis plutonic episode

Benford and others, 2010
Bickford and others, 1981
Criss and Fleck, 1987
Fleck and Criss, 2004
Foster and Fanning, 1997
Gaschnig and others, 2010
Giorgis and others, 2008
House and others, 2002
Janecke and Snee, 1993
Manduca and others, 1993
Snee and others, 1995
Toth and Stacey, 1992
Unruh and others, 2008

Blue Mountains island-arc complex

Kurz, 2010
Kurz and others, 2011
Lee, 2004
Lund and Snee, 1988
McClelland and Oldow, 2007
Snee and others, 2007
Unruh and others, 2008

Paleozoic intrusive rocks

Lund and others, 2010

Neoproterozoic intrusive and volcanic rocks

Alexander and others, 2006
Fanning and Link, 2004
Lund and others, 2003

Mesoproterozoic intrusive rocks

Doughty and Chamberlain, 1996
Evans and Zartman, 1990

Proterozoic metasedimentary rocks

Lewis and others, 2007c
Lewis and others, 2010
Link and others, 2007
Lund and others, 2008

Archean and Paleoproterozoic intrusive rocks

Brewer and others, 2008
Doughty and others, 1998
Doughty and Chamberlain, 2007
Lewis and others, 2011
Link and others, 2007
Vervoort and others, 2007

GARNET ISOGRAD

The garnet isograd is shown on the map to indicate the degree of metamorphism across the state. Because isotopic dating of garnet has shown two periods of growth in the Cretaceous (Getty and others, 1993) and several periods of growth in the Proterozoic (Zirakparvar and others, 2010), the isograd is clearly composite. Nevertheless, it provides a useful guide to regional metamorphic grade. In addition to the map sources listed in Figure 1, information on the location of the garnet isograd was obtained from Hamilton (1963), Myers (1982), and Onasch (1987).

CROSS SECTIONS

Six cross sections were drafted at 1:500,000 scale without vertical exaggeration and then scanned and digitized. All are highly schematic. Some map units were combined because they were too thin to show in cross section; these are noted below the appropriate lines of section. In addition to the source maps shown in Figure 1, several publications specific to structural studies helped in constructing these cross sections. These publications and their corresponding sections are listed here:

D-D'-D'' Skipp (1985)
E-E' Royse and others (1975); Royse (1993)
F-F'-F'' Williams and others (1982); Janecke and others (2003); Pogue (2008)

ACKNOWLEDGMENTS

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Discussions with numerous colleagues were invaluable to developing this map. At the Idaho Geological Survey, John Kauffman, Kurt Othberg, William Phillips, Bill Bonnicksen, Virginia Gillerman, and Roy Breckenridge reviewed early versions and assisted with unit designations. Diana Boyack in the Idaho State University Geosciences Digital Mapping Lab helped with innumerable technical issues during the production of the original county-level maps. Also at the Idaho Geological Survey, Jesse Bird and Collette Gantenbein assisted with digitization, and Collette Gantenbein and Dean Garwood created the base map and shaded relief. Jane Freed and Collette Gantenbein produced the colors and patterns for the map. Their efforts, and remarkable patience, are greatly appreciated. Technical reviews by Russell Burmester, Susanne Janecke, Betty Skipp, Karen Lund, David Rodgers, and Basil Tikoff improved the map in many ways. Editorial reviews by John Kauffman and Roger Stewart are also much appreciated.

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